

A student rubs a plastic rod with fur giving the rod a charge of -1×10^{-6} C. The number of electrons gained by the rod is most nearly

- ☐ A. 2×10^{-13} [10%]
☐ B. 2×10^{13} [29%]
☒ C. 6×10^{12} [51%]
☐ D. 6×10^{18} [8%]

Correct

51% answered correctly

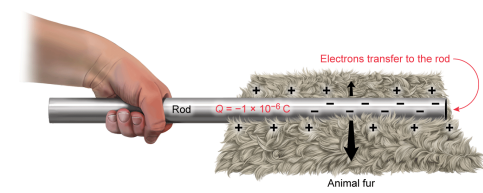
Explanation:

Calculating the number of electrons from electric charge gained by a rod rubbed on fur

$$n = \frac{Q}{e}$$

- n Number of electrons
- Q Electric charge
- e Charge of an electron

$$n = \frac{-1 \times 10^{-6} \text{ C}}{-1.6 \times 10^{-19} \text{ C/e}} = 6.24 \times 10^{12} \approx 6 \times 10^{12}$$



Electric charge is quantized, and an **electron** carries the smallest unit of charge that can be isolated, which is the **elementary charge** $e = -1.6 \times 10^{-19}$ C. A neutral object has an *equal* amount of **positive and negative charge**. When rubbing two objects, electrons can transfer from one object to the other as electric charge is neither created nor destroyed. Hence, the charge Q on an object is some integer multiple n of the elementary charge:

$$Q = ne$$

Thus, the number of electrons on a charged object is:

$$n = \frac{Q}{e}$$

In this question, the plastic rod gains a charge of -1×10^{-6} C after being rubbed with fur. Therefore, the number of electrons that make up this charge is calculated as follows:

$$n = \frac{Q}{e} = \frac{-1 \times 10^{-6} \text{ C}}{-1.6 \times 10^{-19} \text{ C/e}} = 6.24 \times 10^{12} \approx 6 \times 10^{12} \text{ electrons}$$

(Choice A) 2×10^{-13} results from incorrectly dividing e by Q . The number of electrons equals Q divided by e .

(Choice B) 2×10^{13} results from mistakenly subtracting the power of exponents when dividing e by Q .

(Choice D) 6×10^{18} is the reciprocal of the value of e (ie, $\frac{1}{e}$). However, the number of electrons is equal to the charge on the object Q divided by e .

Educational objective:

The charge on an electron is quantized and is the smallest amount of charge that can be isolated in nature. Number of electrons on an object equals the total charge divided by the charge of the electron.

An experiment consists of two positively charged spheres that are set apart at a fixed distance. How will the electrostatic force F_E between the two spheres change if several drops of positively charged oil are added to one of the spheres?

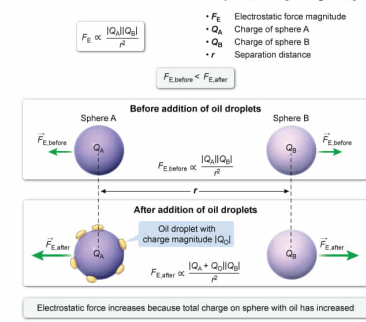
- ✓ ☒ A. The F_E acting on the sphere without the oil will increase, because the total charge on the sphere with oil has increased. [78%]
- ☐ B. The F_E acting on the sphere without the oil will decrease, because the total charge of the sphere with oil has increased. [9%]
- ☐ C. The F_E acting on the sphere without the oil will not change, because its charge is not changed. [6%]
- ☐ D. The F_E acting on the sphere without the oil will not change, because the distance separation of each sphere is not changed. [5%]

Correct

78% answered correctly

Explanation:

Coulomb's Law and electrostatic force experienced by charged objects



Coulomb's law states that the magnitude of the electric force F_E between two **charged objects** is **directly proportional** to the product of each object's charge magnitude ($|Q_1|$ and $|Q_2|$) and **inversely proportional** to the square of the charge's separation distance r .

$$F_E \propto \frac{|Q_1| \cdot |Q_2|}{r^2}$$

In this question, two charged spheres (A and B) are separated by at a fixed distance and exert equal magnitude but oppositely directed electrostatic forces on each other:

$$F_{E, \text{Before}} \propto \frac{|Q_A| \cdot |Q_B|}{r^2}$$

When several positively charged drops of oil (charge Q_o) are subsequently added to one of the spheres (eg, sphere A), the overall magnitude of charge will increase on that sphere. Consequently, the electrostatic force on sphere B will increase according to Coulomb's Law:

$$F_{E, \text{After}} \propto \frac{|Q_A + Q_O| \cdot |Q_B|}{r^2}$$

$$F_{E, \text{After}} > F_{E, \text{Before}}$$

Therefore, the electrostatic force acting on the sphere without the oil (sphere B) will increase because the total charge on the sphere with the oil (sphere A) droplets has increased.

(Choice B) The electrostatic force between the spheres can only decrease if each sphere's charge is decreased or if the spheres' separation distance is increased.

(Choice C) The electrostatic force exists as a mutual force between the two spheres. Thus, an increase in charge on either sphere results in increasing the mutual electrostatic force on both spheres.

(Choice D) The spheres' separation distance remains constant. However, the charge magnitude is increased on one of the spheres, so the electrostatic force increases according to Coulomb's Law.

Educational objective:

Coulomb's Law states that the electrostatic force between two charged objects is directly proportional to the product of charge magnitudes. Increasing the charge on one object increases the electrostatic force on the other object.

Time spent: 0 sec
 Subject:
 Physics

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 System:
 Electrostatics and Circuits